Determinants of Technological Breadth and Strategic Orientation in Hybrid Bonding Patents: A Count and Discrete-Choice Analysis

Technology Management Quantitative Analysis — Final Report

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Abstract

Hybrid bonding has become the bottleneck process for sub-micron interconnect in HBM and 3D logic. The technology itself has been studied closely, but in-depth analysis of its patents remains scarce. This paper treats the patent record as data for econometric estimation. Working from 928 hybrid-bonding applications (5,277 application–CPC records) drawn from EPO PATSTAT, I model two patent-level outcomes. The first is technological breadth, the count of distinct CPC subclasses a patent spans; the second is strategic orientation, whether a patent reaches into front-end fabrication (CPC H01L21) rather than staying at the bonding and interconnect level (H01L24). Because breadth is a count, I begin with OLS, show through diagnostics that it is mis-specified, and move to Poisson and then negative binomial. The OLS residuals are heteroskedastic (Breusch–Pagan $p < 0.001$) and the Poisson is overdispersed (ln-alpha = −1.316, alpha ≈ 0.27, $p < 0.001$), so the negative binomial is the preferred model and has the lowest AIC (4883 against 5477 for Poisson). Breadth widens with filing year and is about 21% narrower for filings at the Chinese office than at the U.S. office. Strategic orientation is then estimated with binary and multinomial logit: broader patents are far more likely to claim fabrication-process technology, the process share has fallen over time, and Chinese-office filings are markedly more process-oriented while Japanese-office filings almost never take a process-only form.

Keywords: hybrid bonding; patent analysis; technological breadth; negative binomial regression; multinomial logit

1. Introduction

The slowdown of dimensional scaling has pushed the semiconductor industry to look for performance gains after the transistor is fabricated rather than only inside it. Heterogeneous integration — partitioning a die into chiplets and reassembling them in the package — is the response, and its success depends on how densely and cleanly those chiplets can be wired together. Hybrid bonding, which joins copper pads and the surrounding oxide directly without solder bumps, pushes interconnect pitch below one micron, and it is now drawing attention as a general-purpose technology (GPT) for the future of semiconductors — reaching across logic, memory, CMOS image sensors, power devices such as GaN and SiC, and even MicroLED.

This study focuses on two attributes of hybrid bonding patents. One is how broad a patent is — the number of distinct technology classes it touches, which proxies for the scope it seeks to protect. The other is where it sits in the value chain — whether it claims front-end fabrication or stays at the bonding and assembly level. The first is a count; the second is a category.

In substance, the question is whether hybrid bonding, as a back-end process that now reaches into front-end territory, is genuinely blurring the line between front- and back-end as the industry evolves. The study answers three questions with models suited to each outcome. Does breadth widen over time and differ by filing jurisdiction? Does the probability that a patent claims fabrication-process technology depend on its breadth, filing year, and jurisdiction? And how is the three-way strategy type — assembly-only, integrated, or process-only — distributed across jurisdictions? The contribution is not a new index but a demonstration that scope and strategic position can be estimated, with the model chosen to fit the statistical nature of each outcome rather than forced into a single linear specification.

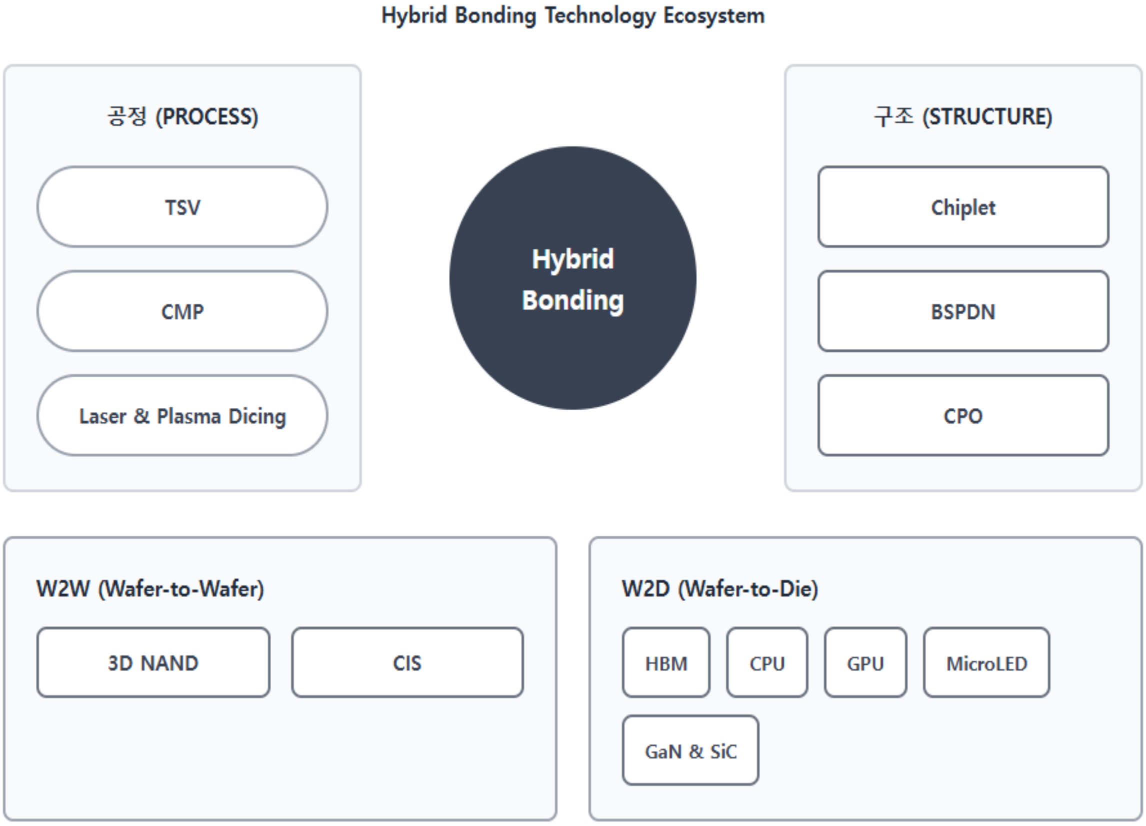

Figure 1. Hybrid bonding technology ecosystem: process, structure, and applications (products).

2. Data and Variables

In the PATSTAT data, each row pairs one application with one CPC classification symbol, giving 5,277 rows for 928 unique applications filed between 1968 and 2024. Every symbol sits under CPC class H01L (semiconductor devices), and within it the records divide between H01L21, which covers fabrication and processing, and H01L24, which covers bonding and interconnect.

These two areas capture the character of the technology. In practice the process divides into wafer-to-wafer (W2W) bonding, which joins whole wafers, and wafer-to-die (W2D) bonding, which attaches individual dies to a wafer; fabrication (H01L21) and bonding/interconnect (H01L24) are realized together. After collapsing to the application level I constructed the variables in Table 1.

Table 1. Variable definitions and construction.

Variable	Definition	Type
breadth	Number of distinct CPC subclass symbols on the application (technological scope)	Count, 1–21
has_process	1 if the application carries any H01L21 (fabrication-process) symbol	Binary
tech_cat	Assembly-only (H01L24 only) / Integrated (both) / Process-only (H01L21 only)	Categorical, 3
ccode	Filing office: US, CN, KR, TW, JP, EP, WO, Other (US = reference)	Categorical, 8

yc	Filing year measured as years since the first filing in the sample	Continuous

Breadth averages 5.69 distinct subclasses (s.d. 4.07, range 1–21). By volume the U.S. office leads with 382 applications, ahead of China (175), the PCT/WO route (113), the EPO (83), Taiwan (75), Korea (58) and Japan (26). By strategy, 425 applications are integrated, 338 assembly-only and 165 process-only; in all, 63.6% carry at least one fabrication-process symbol. Figure 2 shows the filing-year profile. Activity is negligible before 2010 and climbs steeply through the late 2010s, peaking around 2022; the fall in 2023–24 reflects the lag before recent applications are fully published rather than a real decline. The steep recent build-up can be read as a signal that the technology is approaching volume production.

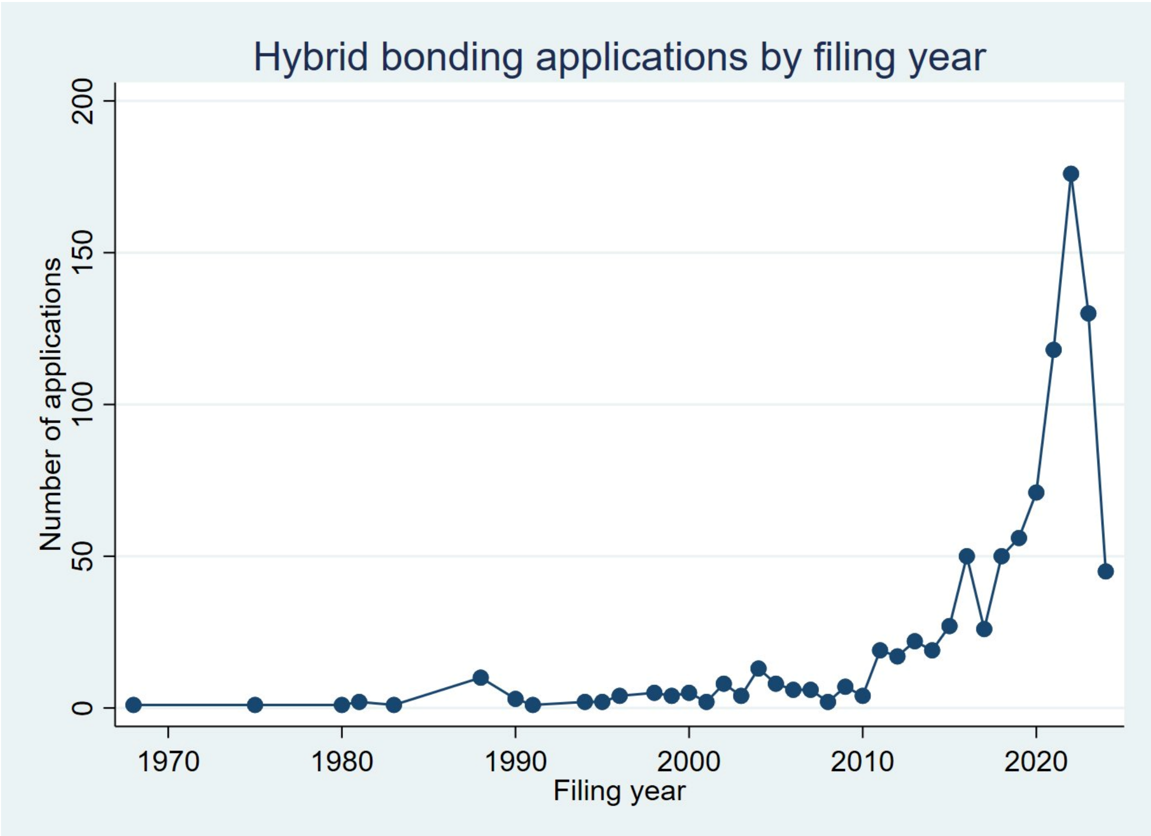

Figure 2. Hybrid bonding applications by filing year. The 2023–24 dip is a publication-lag artifact.

One caveat needs stating up front. The jurisdiction variable is the office where the application was filed, not the nationality of the applicant, and WO and EP are international and regional routes rather than countries. Differences across this variable therefore reflect filing and protection strategy as much as where the invention originated, and I read them in that light throughout.

3. Empirical Strategy

The baseline regresses breadth on the filing-year trend and a full set of jurisdiction dummies with the U.S. office as the reference. I then test the OLS residuals for heteroskedasticity (Breusch–Pagan) and inspect the residual-versus-fitted plot, fit a Poisson model, and check for overdispersion through the estimated dispersion and the likelihood-ratio test on the negative binomial's over-dispersion parameter. Coefficients in the preferred count model are read as incidence-rate ratios.

Strategic orientation is modeled as discrete choice. A binary logit predicts has process from breadth, the year trend and jurisdiction, reported as odds ratios and assessed with the area under the ROC curve. A multinomial logit then takes the three-way strategy type with assembly-only as the base category, reported as relative-risk ratios, so that the integrated and process-only branches can be read against the same reference.

4. Results

4.1 Technological breadth

Table 2 collects the breadth models. In OLS the year trend is positive and significant (0.070, $p < 0.01$): patents have grown broader as the field matured, though the model explains little of the variation ($R^2 = 0.033$). The residual diagnostics are the point of interest. Breusch–Pagan rejects constant variance ($p < 0.001$), and the residual-versus-fitted plot in Figure 3 shows the diagonal striping and fan shape that a count outcome produces — variance rising with the fitted value and residuals bounded below. Robust standard errors change little, which tells us the problem is the functional form, not just the errors.

*Table 2. Determinants of technological breadth. US office is the reference; standard errors in parentheses; * p<0.10, ** p<0.05, *** p<0.01.*

	OLS	OLS (robust)	Poisson	Neg. Binomial
Filing year	0.070***	0.070***	0.013***	0.014***
	(0.019)	(0.020)	(0.002)	(0.003)
China (CN)	−1.321***	−1.321***	−0.241***	−0.242***
	(0.370)	(0.330)	(0.040)	(0.062)
Korea (KR)	0.089	0.089	0.017	0.017
	(0.570)	(0.630)	(0.059)	(0.094)
Taiwan (TW)	0.102	0.102	0.016	0.021
	(0.510)	(0.544)	(0.051)	(0.083)
Japan (JP)	1.737**	1.737	0.280***	0.239*
	(0.845)	(1.328)	(0.080)	(0.133)
EPO (EP)	−0.032	−0.032	−0.005	−0.007
	(0.487)	(0.453)	(0.050)	(0.080)
PCT (WO)	−1.071**	−1.071***	−0.193***	−0.191***
	(0.432)	(0.399)	(0.047)	(0.073)
Other	0.124	0.124	0.024	0.056
	(1.029)	(1.556)	(0.106)	(0.171)
Constant	2.506***	2.506**	1.143***	1.069***
	(0.944)	(1.046)	(0.107)	(0.175)
ln(alpha)				−1.316***
				(0.076)
N	928	928	928	928
R-sq.	0.033	0.033	—	—
AIC	5224.9	5224.9	5477.2	4883.0

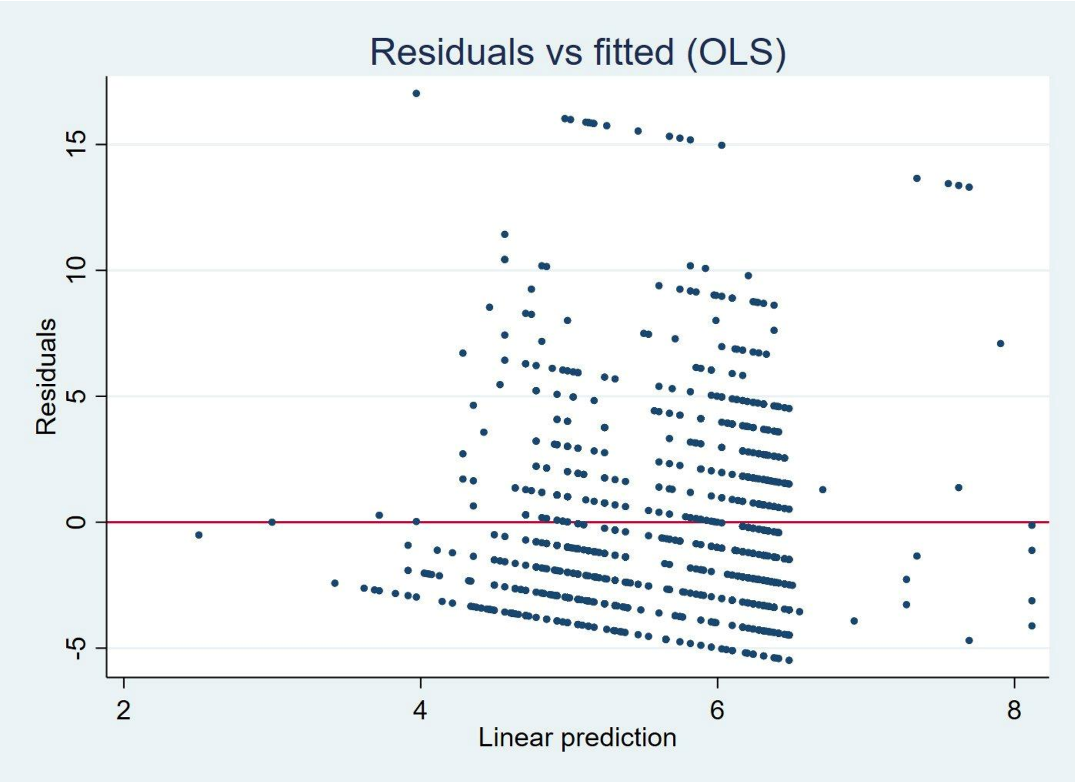

Figure 3. OLS residuals versus fitted values. The banding and funnel shape signal a count process and heteroskedasticity.

Moving to Poisson and then negative binomial confirms this. The negative binomial's over-dispersion parameter is large and precisely estimated (ln-alpha = −1.316, alpha ≈ 0.27), the likelihood-ratio test against Poisson is decisive, and AIC drops from 5477 under Poisson to 4883 under negative binomial. I therefore read the negative binomial as the working model. The year trend survives (IRR ≈ 1.014 per year). Against the U.S. baseline, filings at the Chinese office are about 21% narrower (IRR 0.785, $p < 0.01$) and PCT/WO filings about 17% narrower (IRR 0.826, $p < 0.01$), while Japanese-office filings are the broadest in the sample (IRR ≈ 1.27, $p < 0.10$). Korea, Taiwan and the EPO do not separate from the U.S. baseline. Figure 4 plots the predicted breadth by jurisdiction that the model implies.

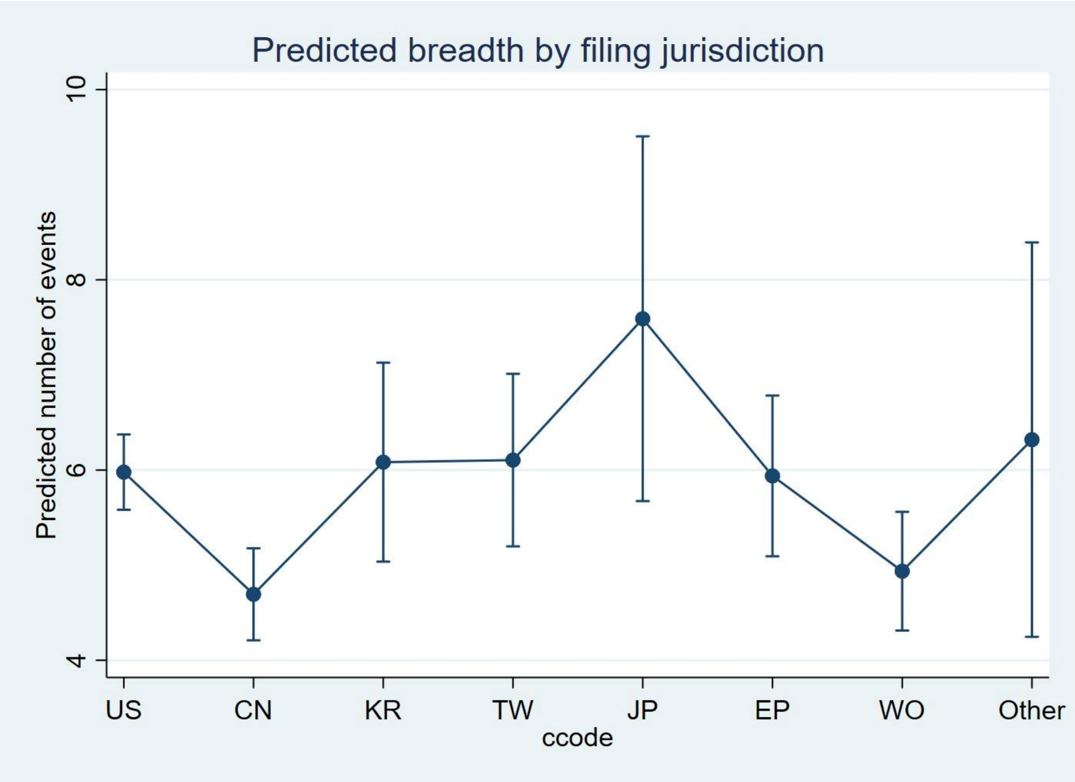

Figure 4. Predicted technological breadth by filing jurisdiction (negative binomial, other covariates at their means).

4.2 Strategic orientation

Table 3 reports the binary logit for whether a patent claims fabrication-process technology. Breadth is the dominant predictor: each additional CPC subclass raises the odds of a process claim by about 27% (OR 1.267, $p < 0.01$), which fits the idea that patents reaching back into front-end steps are also the ones that span more classes. The year trend runs the other way (OR 0.946 per year, $p < 0.01$) — the share of process-oriented filings has fallen as bonding- and assembly-level invention has taken over the field. Among jurisdictions only the Chinese office stands out, at 65% higher odds of a process claim than the U.S. office (OR 1.650, $p < 0.05$). The model discriminates acceptably, with an area under the ROC curve of 0.707 (Figure 5).

*Table 3. Binary logit of process orientation (has_process). Odds ratios; standard errors in parentheses; * p<0.10, ** p<0.05, *** p<0.01.*

	Odds ratio	**Std. err.**
Filing year	0.946***	(0.011)
Breadth	1.267***	(0.034)
China (CN)	1.650**	(0.338)
Korea (KR)	1.430	(0.467)
Taiwan (TW)	1.555	(0.451)
Japan (JP)	0.483	(0.233)
EPO (EP)	1.170	(0.312)
PCT (WO)	1.398	(0.329)
Other	2.132	(1.307)
N = 928 · log-lik = −547.6 · AIC = 1115.1 · AUC = 0.707		

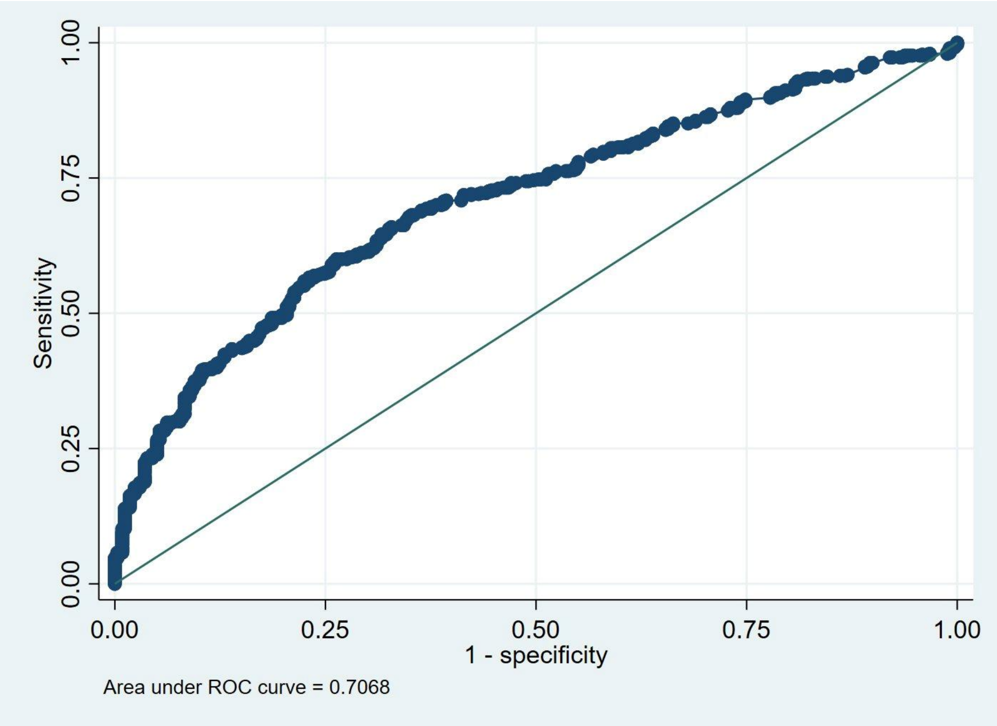

Figure 5. ROC curve for the process-orientation logit (AUC = 0.707).

The multinomial logit in Table 4 separates the two ways a patent can be a pure play. Read against assembly-only, breadth pushes a patent toward the integrated type (RRR 1.403, $p < 0.01$) and away from process-only (RRR 0.905, $p < 0.05$): broad patents straddle both layers, narrow ones specialize. The clearest contrasts are by jurisdiction in the process-only branch. The Chinese office is far more likely to host a process-only patent than the U.S. office (RRR 2.687, $p < 0.01$), with Taiwan and the PCT route also elevated, whereas the Japanese office almost never does (RRR 0.086, $p < 0.01$). The year trend again pulls down the process-only share (RRR 0.913, $p < 0.01$). Figure 6 shows the predicted probability of a process-only strategy by jurisdiction implied by the model.

*Table 4. Multinomial logit of strategy type (relative-risk ratios, base = assembly-only). Standard errors in parentheses; * p<0.10, ** p<0.05, *** p<0.01.*

	Integrated	(SE)	Process-only	(SE)
Filing year	0.993	(0.015)	0.913***	(0.013)
Breadth	1.403***	(0.043)	0.905**	(0.044)
China (CN)	1.205	(0.280)	2.687***	(0.740)
Korea (KR)	1.166	(0.444)	1.906	(0.799)
Taiwan (TW)	1.312	(0.422)	2.102*	(0.857)
Japan (JP)	1.046	(0.583)	0.086***	(0.070)
EPO (EP)	1.307	(0.380)	0.858	(0.376)
PCT (WO)	1.126	(0.302)	1.997**	(0.645)
Other	2.326	(1.585)	1.901	(1.480)
N = 928 · log-lik = −774.5 · AIC = 1589.0 · base = Assembly-only				

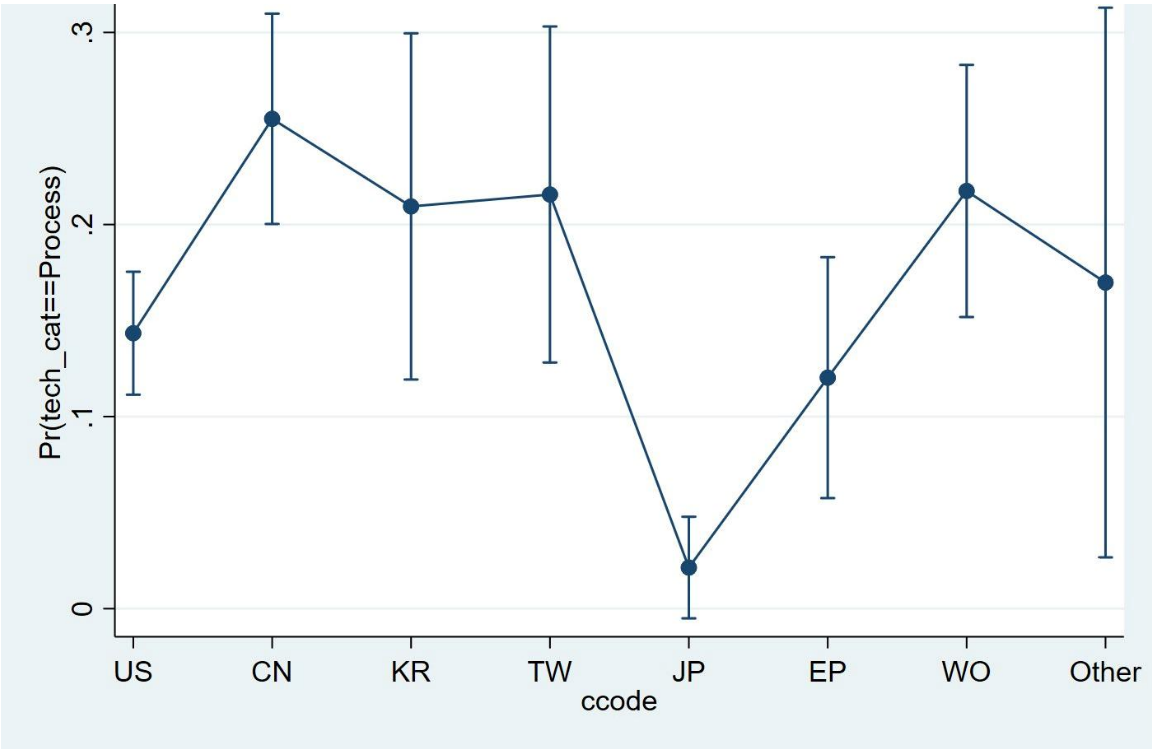

Figure 6. Predicted probability of a process-only strategy by filing jurisdiction (multinomial logit).

5. Discussion

The jurisdiction pattern is the part worth dwelling on, with the filing-office caveat kept in mind. Chinese-office filings are narrower in scope and the most likely to be process-only; Japanese-office filings are the broadest and essentially never process-only. These are recognizable strategies. A narrow, process-focused filing profile is what one expects from a catch-up posture that concentrates on internalizing specific fabrication steps — copper surface preparation, low-temperature anneal, dicing — rather than claiming whole integrated stacks; it points to a focus on semiconductor fabrication itself rather than on back-end packaging. The Japanese profile, broad and integration-oriented, is consistent with the equipment and materials suppliers who patent across the full process flow. Korea and Taiwan sit near the U.S. baseline, which is plausible for the memory makers and the foundry that compete head-on with U.S. assignees.

The time trends point the same way for both outcomes: breadth rises and the process share falls. Together they describe a field maturing out of its foundational fabrication patents and into broader bonding- and integration-level claims — the same shift, in patent terms, that the industry has lived through as hybrid bonding moved from wafer-to-wafer use in image sensors and 3D NAND toward die-to-wafer integration for HBM and advanced logic.

6. Conclusion and Limitations

Estimated rather than merely counted, the hybrid-bonding patent record shows technological breadth widening over time and varying by filing jurisdiction, and a strategic orientation that has drifted from fabrication toward assembly while remaining sharply different across offices. Methodologically the paper makes the case that the model has to follow the outcome: a count belongs in a negative binomial once overdispersion is shown, and a categorical strategy belongs in a multinomial logit.

Three limitations bound these claims. The jurisdiction variable is the filing office, not applicant nationality, so cross-office differences mix invention origin with protection strategy. The estimates are associational; breadth and strategy are plausibly jointly determined, and with no credible instrument available in the data I do not attempt an instrumental-variables correction and leave causal identification to later work. And without applicant identifiers I cannot add firm fixed effects, which a linked dataset would allow and which would sharpen every comparison made here.

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